

Physics Activities

Fast-Paced Competitions

Here are some fast-paced competitions you can participate in as a high schooler in the US:

- The [F = ma/USAPhO](#) exams are the premier physics competitions in the US. The $F = ma$ exam is a fast-paced multiple choice exam (75 minutes, 25 questions) focusing on tricky mechanics problems, while the USAPhO exam has longer problems which require written solutions (3 hours, 6 questions).
- The [PhysicsBowl](#) is a fast-paced multiple choice exam (45 minutes, 40 questions) at the level of AP Physics 1 and 2, mixed with trivia and history questions.
- The [Science Bowl](#) is an exciting buzzer-based team competition covering all fields of science. You need a team of four, preferably with everybody specialized in one or two subjects, and qualifying for the national competition can be very competitive if you live in a large state. To do well as the team's physics player, you need to be able to solve AP Physics 1 and 2 questions *extremely* quickly, and also have a good deal of general physics knowledge. If you have a lot of general knowledge about science, consider being the science player on a [quiz bowl](#) team.
- Check to see if your state has local science competitions. For example, in New Jersey you can participate in the [New Jersey Science League](#).
- The [Physics Unlimited Premier Competition](#) is a fast-paced short answer exam (90 minutes, 4 questions), which can also be taken internationally. It's written by college physics majors and generally has questions drawn from standard undergraduate textbooks.
- It's important to remember that even if you're a hotshot at high school physics, there's a lot more to learn. For example, try playing [arXiv vs. snarXiv](#) to see if you can even tell the titles of real physics papers from randomly generated ones.

Slower-Paced Competitions

Next, here are some slower-paced competitions, generally open internationally. These tend to have less name recognition, but they're fun and can help build long-term problem solving skills.

- [KoMaL](#) and [FYKOS](#) are Hungarian and Czech contests for high school students which issue a problem set every month. The problems in both tend to be elegant and instructive.
- The team behind FYKOS also runs the team-based [Online Physics Brawl](#) and [Fyziklani](#) competitions, which each last a few hours and have about 50 questions. Unfortunately, you'll have to stay up late to participate if you live in the US.
- The AAPT's journal, [The Physics Teacher](#), publishes a monthly problem under the column "Physics Challenge for Teachers and Students". The difficulty is usually between an $F = ma$ and a USAPhO problem.
- The [Physics Unlimited Explorer Competition](#) is a team-based, two week competition where students work through a series of exercises exploring a topic in undergraduate physics.

- The [High School Mathematical Contest in Modeling](#) and [MathWorks Math Modeling Challenge](#) are team-based, one to two day competitions where students are faced with an open-ended real-world problem and write a report issuing policy recommendations.
- The [USAYPT](#) is a team-based, year-long competition where students investigate open-ended *experimental* physics problems. Out of all the competitions listed here, this one gives students the most freedom by far, but it requires more to get started. To have a shot, your high school needs to already have some sort of physics club, and the resources to get decent lab equipment.
- The [Online Physics Olympiad](#) is a team-based, several day competition written entirely by high schoolers, founded in 2020. It is one of the largest international physics competitions, with hundreds of teams participating. Because it's student written, the problems can be contrived or ambiguous, but they're rapidly improving over time.
- The [Physics Cup](#) is an extremely difficult competition with one question per month, with a hint released per week. The problems are very instructive, and range from applied physics to almost pure mathematics. They always are clearly posed and have unambiguous answers, but they are sometimes so hard that almost nobody can solve them even after four weeks of hints. In theory, all the problems can be solved with only introductory physics. A good knowledge of Euclidean geometry tends to help.
- The [Rudolf Ortway](#) competition is a marathon competition with about 30 questions to be solved in 10 days, requiring undergraduate physics knowledge to solve.

Science Fairs

You can also consider entering a high school science fair, such as ISEF or Siemens, which have great name recognition. Unfortunately, I'm not qualified to give advice on this, since I placed at the bottom of my county's science fair, after being dragged there by a graduation requirement. My comparison of bean growing methods, with everything bought at Home Depot, was no match against the genetic engineering projects other people showed up with, using equipment from university labs.

I remember spending the day forlornly glancing back and forth, from the Excel beanstalk height bar charts on my cardboard poster, to the towering achievements all around me. Gene therapy, antibiotic resistance, cancer immunotherapy – even their poster material was nicer! For quite some time, this made me doubt I could ever be a scientist. I must have been lacking some inherent spark, some essential creativity, to have come up with those projects myself.

Of course, that wasn't right at all, and the reason is that science fairs don't work like the competitions I've listed above. In physics competitions, you can do well by learning physics on your own, or in any decent school, and thinking hard about it. For them, I never needed anything but a dog-eared, \$20 used textbook, but in science fairs, you [need](#) to know [people](#). The reason is that it's almost impossible to come up with and carry out a novel and important physics research project sitting on your own at home, given the immense background required; even the nation's brightest child prodigies are barely at the level of the average new graduate student. The press releases try to make it sound like the projects spring out of a flash of creative inspiration, but in reality, the projects are simply given to students by professors at universities, who guide them through step by step. The students work on subsubtasks like any diligent research assistant would, but the overall direction, and even the choices of subtasks, are the professor's.

Now, most professors won't even work with undergraduates in research, because they lack the background of graduate students. And high schoolers have even less background than undergrads. That means the professors that are willing to serve as research mentors are outnumbered by high school students looking for them by a factor of a hundred to one, or perhaps even a thousand to one. Given those odds, how can you find one?

In practice, there are two options: either cold email many professors and get extremely lucky, or have an "in". For example, you can be born to parents who themselves are professors, so they can hire you in their own labs, or [send you](#) to their friends' labs. Or you could just have very involved parents, who will dig into the system and [fight for you](#). You can go to an elite high school which has a dedicated class that mass-produces winning science fair projects. If all else fails, you can shell out for a few hours of Zoom calls with a past science fair winner, so you can ask them how they did it. These days, they charge [about \\$9,500](#) for the privilege.¹

Even once you have a mentor, your success in a science fair hinges on external factors. When you do high school physics competitions, or research in graduate school, success is driven by internal factors: you decide how many hours to work, which avenues look promising, and what to try next. But a high schooler doing research almost never has the background to operate that way. Success will only be possible if your mentor goes to the trouble of developing a very concrete, well-posed subproblem for you to work on, while they handle all the creative direction and the details that require advanced knowledge.² The fact that this sounds like a pretty bad deal for the mentor (whose student will soon go off to college, never to return) is precisely why you need an "in" to get one, which in turn is highly correlated with your pedigree.

In my opinion, the best childhood activities are those directed as much as possible by the kids themselves. The Online Physics Olympiad mentioned above is a great example: in a year where most physics competitions were cancelled, high school students wrote, publicized, ran, and graded a massive competition on their own, with zero budget. By contrast, a science fair is by definition an exercise in appealing to adults, who craft the projects, pay for the equipment, run the fair, hear the presentations, and judge the winners. And because there is so much adult involvement, and the stakes in prestige are so high, the essential playfulness is lost. The mood is serious, or perhaps even farcical. In a scene out of *Bosch*, parents desperately throw suitcases of money in every direction, in the hope that one will unlock the door to an Ivy League college.

Given all this, why do we place value on science fairs at all? Physics Olympiad problems are merely exercises designed to be solvable by high schoolers who learn the subject well. But the end goal of this learning to produce scientists who can produce completely new knowledge, years down the line. A good science fair project, however, purportedly shows that the student can produce new knowledge *today*. What could be a better signal of their potential?³

¹Actually, there's another option: you can completely fake your results, like the [winner](#) of the 2024 ISEF competition did! Of course, this episode was neatly swept under the rug, and you won't find any mention of it in the glossy press releases. But it shows that even at the highest levels, science fairs mainly test that you *sound* like you know things. It's a subtle effect, but I think promoting this kind of behavior harms science as a whole.

²Some of my comments are specific to physics. For example, research projects in the social sciences, or in certain areas of mathematics like combinatorics, require substantially less background. Here, a high schooler really could do novel, interesting work reasonably independently. And if you can tinker, you can make good stuff with relatively cheap electronics. On the other hand, for biology and chemistry, university mentorship and tight guidance is even more necessary than in physics. You need the expensive equipment in university labs, and no professor is just going to let you mess with it. If you're there, they'll be giving you very explicit instructions so you don't waste their money.

³The reason this feels so compelling is what Cal Newport calls the [failed simulation effect](#). If some achievement looks impressive, and you can't see the steps one would take to get it, you naturally conclude that it must spring from some exceptional genius. In reality, they usually come from parents' pocketbooks or personal contacts. For

That’s the narrative, anyway, but I think an equally compelling argument could be made in the opposite direction. What are the traits of good researchers? There are many, but they certainly include a deep understanding of one subfield, a broad knowledge of all of physics, and the ability to make insights into difficult problems they’ve never seen before. These underlying skills are precisely what is trained by physics Olympiad problems, many of which illustrate the key insights behind real breakthroughs in physics. If you really care about research, I think it’s better to spend your high school years building the foundation that will someday make you a good researcher, rather than impressing physics-illiterate adults with the superficial appearance of the final product.

I have nothing against science fair participants, and I’m friends with a bunch of national science fair winners, many of whom also did Olympiads. The national winners are smart and dedicated. But if you want to stand a chance to *qualify* for the national science fairs, and you don’t already know how the game is played, you’re not going to find anything useful here, or in any book. You’re not going to get anywhere with \$20 or a handful of beans. You need to know the right people.

Programs

Okay, so how do you find the right people? I’ve gathered some advice from friends who have won national science fairs. First, you can look at organized programs.

- If you are part of an underrepresented minority group, there are many well-organized and well-funded summer programs which will take care of the process for you and let you meet a group of similar students, such as [SAMS](#), [MITES](#), [SMASH](#), [YRP](#), [SSRP](#), and many more.
- If you’re not part of such a group, you should avoid most of the well-promoted programs out there. As one of many terrible examples, the [Summer STEM Institute](#) is a Zoom-based program run by undergraduates with no experience formulating research questions (as mentioned above, even national winners are simply told what to do by their mentors). Programs like these often promise you’ll get a publishable paper after just a few hour-long meetings with a random college student, which is completely absurd if you know anything about research.⁴ They also cost up to \$10,000, which means you need to pay over \$1,000 per hour of actual mentorship! Make no mistake: these programs are just a trap for desperate and unwise parents, run by deeply cynical kids looking to make a quick buck. They embody everything wrong with the American science fair and college admissions process.
- Anyway, if you’re aiming for pure math research, one excellent program is [MIT PRIMES](#), which matches high school students with qualified mentors for free. Admissions are relatively transparent, requiring the solution of a problem set; Olympiad qualifications help. Also see [this advice](#) from the Euler Circle.

example, when you hear that, say, a 12 year old kid built a nuclear reactor in their garage, you should understand that (1) a very inclusive definition of “nuclear reactor” is being used, with the products more closely resembling smoke detectors and [bananas](#) than power plants, (2) there are established hobbyist communities for this kind of thing, and step-by-step instructions, (3) the main reason few kids do it is that the materials cost tens of thousands of dollars and require adult supervision to put together – aspects all carefully scrubbed out of the press release. (And who do you think wrote the press release?) Of course I don’t mean to condemn these kinds of activities; they’re probably a great way for rich parents to bond with their kids. You just shouldn’t feel bad if you didn’t do the same as a kid.

⁴Some of the larger programs solve this problem by [running their own journals](#), which will accept just about anything, even essays at the level of a high school paper. Their standards are so low that the results won’t succeed even in regional science fairs, and I have no doubt even the slowest admissions officer will soon understand that too.

- For general science research, the best program is [RSI](#), which is also free. RSI is an extremely selective program, and one of its explicit goals is to allow its students to go to any selective college they want, which is achieved by very strong recommendation letters. However, because of its prestige, if you don't come from a very small state, you usually need to *already* have research experience to stand a chance of admission.
- There are also great physics summer programs that are more about learning than research.
 - [ISSYP](#) is a nearly free program covering topics in theoretical physics, held at the beautiful Perimeter Institute. [QSYS](#) is a free program run by the nearby University of Waterloo, focusing on quantum computing.
 - [SPARC](#) is a free program which covers “applied qualitative thinking”, which means an array of fun topics such as game theory, economics, and cognitive science. Though it doesn't have much to do with physics, it attracts many people in the Olympiad community and is a lot of fun, since there's plenty of free time for discussions. A major part of the application is a set of interesting, open-ended questions. The [Atlas Fellowship](#) is another intriguing program along the same lines, funded by the same community.
 - [SSP](#) and [YSPA](#) are relatively expensive but fun programs for high school juniors focused on astrophysics; students hang out in observatories overnight taking real data.

Zoom-based camps are much less fun than in-person camps, because all the informal socialization is lost, leaving only dry lectures. I wouldn't ever recommend paying for a Zoom-based camp.

- I also wouldn't recommend paying for a summer camp at a prestigious university, as these tend to be absurdly overpriced. For example, Stanford charges \$20,000 for two months of mediocre, watered down courses, which often aren't even taught by actual professors. Because admission is based on your ability to pay, rather than merit, attending such a program has no benefit for college admissions. You can get the same knowledge elsewhere for a hundred times cheaper.

Starting Research

While organized programs are fun, outside of pure math, most science fair winners don't get their projects from them. The main method is reaching out to professors directly.

- The hardest stumbling block, at least for me, was realizing this is a sensible thing to try. As mentioned above, it's highly unlikely that you could do *anything* more efficiently than a professor or even a graduate student, since you are missing years of knowledge. But professors really do choose to mentor high school students sometimes, for a variety of reasons: your time is free, you could become more useful if you stick around, and they might like you for some reason.
- If you have the privilege of personally knowing lots of professors already, you're very lucky; stop reading, and just ask them the next time you bump into them at the country club. Or, if you have a spare \$10,000, you can sign up with one of the highly profitable “college consulting” startups which mass-produce these projects. If you're part of the 99%, you won't be able to do either, but these are the most common paths to high school research awards. That's life.
- Otherwise, to get started, browse the faculty listings at nearby universities. Read their biographies and websites and see what sparks your interest. Exclude any professors that are no longer taking students, such as emeriti. Consider friendly faculty who care about mentoring; a

professor happy to work with undergraduates, and who publicly encourages them to apply, is much more likely to be willing to work with high school students.

- Don't be overly picky at this stage. For example, don't fall into the trap of insisting on a string theory project because you once saw a Michio Kaku video – you won't get anywhere that way. Also don't laser focus on the highest ranked university within a thousand miles. Cutting-edge research is done at hundreds of universities in the United States. The most important factor is that the university be close enough for you to regularly show up.
- Pick several professors to send a cold email to. Don't be demanding; be aware that you're asking for a big favor. Also be aware that professors typically skim through hundreds of emails a day. Your first email should briefly convey the following information:
 - Your interest in physics. Don't try to exaggerate here, because it will be obvious if you do. For example, if you've learned some quantum mechanics by reading a textbook, that's a great thing to mention. But don't say you've been dying to work on superconducting spintronics, or whatever it is the professor works on, since infancy. A decent number of professors have heard about Olympiads, so mentioning your awards can help.
 - Your ability to code. In order for a high school student to be helpful at all, the professor needs to be able to find a specific, reasonably sized task for them to do, requiring less technical background. In the vast majority of cases, this means you'll end up coding, which is an essential part of all fields of physics. If you have experience coding, even if it wasn't in physics, you should mention it.
 - The amount of time you can commit. Summer tends to be a good time to start, because you'll be free almost all the time. Don't put a hard time limit (e.g. "I need a result within 3 months for this science fair") because research doesn't work that way. Most decent projects take a year or more.

No matter what your qualifications are, the vast majority of requests will be declined; that's normal. Remember, you only need one!

- Here are some points of email etiquette:
 - In the United States, titles generally don't matter as much as in Europe or Asia, but it's still best to address professors as Prof. Lastname and postdocs as Dr. Lastname in a first email. If they reply to you and sign off with their first name, feel free to use it.
 - Sometimes, high school students have elaborate email signatures detailing their class year, address, school, favorite color, and more. Only administrators use these in real life. Just sign off with "Thanks", followed by your name.
 - Don't send anything that demands substantial energy for the professor to parse, such as a multi-page resume with every minor honor you've ever gotten.
 - If you don't get a response in a week, your email could have just fallen down in their inbox, so you can send a single reminder. Don't send multiple reminders and don't be annoying.
- Remember that research takes a long time, and don't obsess about jumping to a final result for the science fair. You'll probably need weeks of reading and listening just to figure out the basics of what your problem is about. Sometimes, months of work will yield nothing at all!

If you stick around in your research group, you'll find that the real benefit of mentorship has nothing to do with winning the science fair. You'll gain a much deeper appreciation of how science works by being part of a real scientific community. If you work hard and show that you can be a useful part of this community, your mentor will be able to say a lot more about your abilities than a high school teacher ever could.